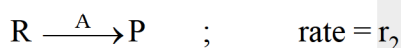
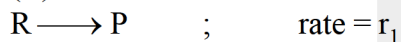


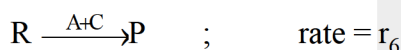
HINTS & SOLUTION

Q.1 (C)

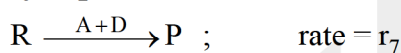
Q.2 (D)



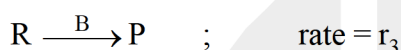
$r_2 > r_1 \Rightarrow A$ is catalyst



$r_6 > r_2 \Rightarrow C$ is catalyst promoter of 'A'



$r_7 < r_2 \Rightarrow D$ is catalytic poison of A



$r_3 < r_2 \Rightarrow B$ is inhibitor

Q.3 (B)

$$\frac{x}{m} = Kp^{1/n}$$

$$\frac{0.2}{0.5} = \frac{K(4)^{1/n}}{K(25)^{1/n}}$$

$$\frac{2}{5} = \left(\frac{2}{5}\right)^{2/n}$$

$$\frac{2}{n} = 1$$

$$n = 2$$

$$\frac{x}{m} = Kp^{1/n}$$

$$0.2 = K(4)^{1/2}$$

$$K = 0.1$$

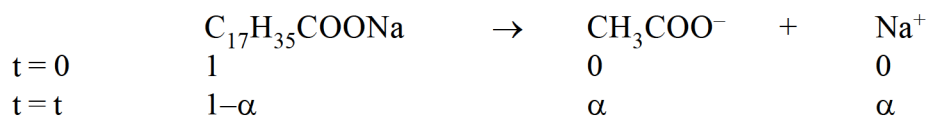
$$\frac{x}{m} = 0.1 \times (36)^{1/2} = 0.6$$

$$\frac{x}{m} = 0.6 \text{ gm}$$

$$\text{Moles} = \frac{0.6}{28} = \frac{3}{140}$$

Q.4 (C)
 $\pi = iCRT$

$$\frac{\pi}{RT} = iC$$



$$i = \frac{1 + \alpha}{1} = 2$$

$$\pi_1 = 0.02$$

$$\pi_2 = 0.4$$

$$\pi_3 = 0.6 < 0.058$$

So here at this point micelle formation has taken place and colloid has been formed. We can say at 0.04 M concentrate mixture will be heterogeneous.

Q.5 (B)
 $H_2(g)$ get adsorbed on transition metals to form metal hydride. It is an example of chemisorption so enthalpy change will be in the range of 80–240 kJ/mole

Q.6 (D)

Q.7 (C)

Q.8 (A)

Q.9 (B)

Q.10 (C)

Q.11 (A)

Q.12 (C)

Q.13 (C)

Q.14 (B)

Q.15 (A)

Q.16 (A)

Q.17 (A)

Q.18 (B)

Q.19 (B)



Q.20 (D)

Q.21 (B)

Q.22 (B)

Q.23 (C)

Q.24 (D)

Q.25 (D)

Q.26 (D)

Q.27 (B)

Q.28 (B)

Q.29 (B)

Q.30 (B)

Q.31 (A)

Q.32 (B)

Q.33 (C)

Q.34 (D)

Q.35 (B) (D)

Q.36 (A)

Q.37 (A) (B) (C)

S_1, S_4 are true statements.

S_2, S_3, S_5, S_6 are false statements.

Q.38 (A) (B) (D)

Extent of adsorption of O_2 is more than H_2 at same temperature.

$\therefore$ y - intercept of $\log \left(\frac{x}{m} \right)$ v/s $\log P$ i.e. $\log k$

$$\left[\log \left(\frac{x}{m} \right) = \frac{1}{n} \log P + \log K \right]$$

will be greater for O_2

$\Rightarrow$ For adsorption $\Delta H = -ve$ (Enthalpy driven process)

$\Delta S = -ve$

$\rightarrow$ Adsorption isobars keeps on decreasing with increase in temperature if there is only vanderwaal's interactions between adsorbent and adsorbate. (Since $\Delta H = -ve$)



Q.39 (A)(C)

Q.40 (A)(B)

Q.41 (A)(D)

Q.42 (A)(B)

Q.43 (A)(B)(D)

Q.44 (A)(D)

Q.45 (B)(C)

Q.46 (B)(C)

Q.47 (B)(D)

Q.48 (A)(C)

Q.49 (B)(C)

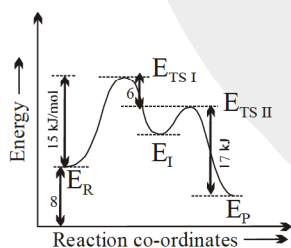
Q.50 (i) – (b) (ii) – (c) (iii) – (d) (iv) – (a)

Q.51 (i) – (d) (ii) – (c) (iii) – (a) (iv) – (b)

Q.52 (i) – (b) (ii) – (c) (iii) – (d) (iv) – (a)

Q.53 (i) – (c) (ii) – (d) (iii) – (b) (iv) – (a)

Q.54 0017



From diagram $E_{TS II} - E_P = 17 \text{ kJ/mole}$

Q.55 0090

$$\left(\log \frac{x}{m} - 1.5\right) = \frac{1}{2}(\log P - 1)$$

$$2\log \frac{x}{m} - 3 = \log P - 1$$

$$\log \frac{x}{m} = [\log P + 2] \frac{1}{2}$$

$$= \frac{1}{2} [\log(81) + \log 100]$$

$$\log \frac{x}{m} = \log (8100)^{1/2}$$

$$\frac{x}{m} = 90$$

Q.56 It is a 4% solution of nitrocellulose in a mixture of alcohol and ether.

Q.57 Animal hide is colloidal in nature and has positively charged particles. When it is soaked in tanin which has negatively charged colloidal particles, it results in mutual coagulation taking place.

Q.58 The hydrogen ion concentration at which the colloidal particles are neither positively charged nor negatively charged (i.e. uncharged) is known as isoelectric point of the colloid.

Q.59 50.0

Q.60 25.0

Q.61 2.47 moles

Q.62 0.4 g

Q.63 (a) Lyophilic (b) Lyophobic (c) Lyophobic (d) Lyophilic

